

Quantum Simulation of the Burgers and Korteweg–de Vries Equations: Carleman–QSVT Linear-System Solving and Variational Natural-Gradient Time Propagation

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We solve two nonlinear partial differential equations on a periodic domain — the viscous Burgers equation and the Korteweg–de Vries (KdV) equation — using two independent quantum algorithms, and we verify every mathematical claim used by either one against classical reference calculations. The first method (Carleman–QSVT) lifts the discretized quadratic ordinary differential equation into a truncated linear system and solves it, at each Crank–Nicolson timestep, with the quantum singular value transformation (QSVT) applied to a block encoding of the linear operator. The second method (variational natural-gradient time propagation) instead encodes the field directly into a parametrized quantum circuit and steps it forward in time by minimizing, via quantum natural gradient descent, the distance to a physically-motivated target state; the circuits used for this involve no more than a single ancilla and no full-ansatz controlled operations. Both pathways use the same periodic finite-difference discretization, closed with an exactly energy-conserving convective term that we derive and prove correct in three lines. This closure is what guarantees, exactly, the bound $\|u(t)\|_2 \leq \|u_0\|_2$ for Burgers and $\|u(t)\|_2 = \|u_0\|_2$ for KdV — the latter requires switching the variational method’s time integrator from forward Euler, which we prove injects energy on every step, to the implicit midpoint rule. We report full resource estimates for both pathways, including a self-contained scaling study of how gate count, circuit depth, and qubit count grow with problem size directly from our own implementation, and discuss honestly the one place where the variational method still depends on an $O(2^n)$ classical computation.

I. INTRODUCTION

We consider two canonical nonlinear partial differential equations (PDEs) on the periodic spatial domain $x \in [0, 1)$:

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad (\text{viscous Burgers}) \quad (1)$$

$$\partial_t u + u \partial_x u + \delta \partial_{xxx} u = 0, \quad (\text{KdV}) \quad (2)$$

where $\nu > 0$ is the viscosity and $\delta > 0$ the dispersion coefficient. Burgers’ equation is parabolic and dissipative: the diffusion term forces the solution to decay, and the exact solution obeys a strict *maximum principle*, $\max_x |u(x, t)| \leq \max_x |u(x, 0)|$, for every $t \geq 0$ (the argument is elementary: at an interior maximum $\partial_x u = 0$, so $\partial_t u = \nu \partial_{xx} u \leq 0$ there). KdV, in contrast, is dispersive and conservative: neither term dissipates, and the L^2 norm is exactly conserved, $\|u(t)\|_2 = \|u(0)\|_2$, because both the dispersive term and the convective term do zero net work (Sec. II A). Both bounds are explicit requirements of the NBQSS 2026 challenge, and both are the physical quantities we designed our discretizations around, rather than treating them as optional sanity checks.

Burgers dynamics are not merely a textbook model equation: they also arise directly as the continuum limit of a real physical system. A Digital Degenerate Cavity Laser (DDCL) — a self-imaging optical resonator whose spatially separated laser sites are coupled with a programmable complex phase via an intracavity spatial light

modulator — realizes an array of Sakaguchi–Kuramoto phase oscillators; in the long-wavelength continuum limit and phase-only regime, the site-phase gradient obeys exactly the one-dimensional viscous Burgers equation, with the effective viscosity set by the real part of the optical coupling and advection by its imaginary part [16]. This gives an independent, non-quantum experimental realization of the same PDE we solve quantum-mechanically below, and underlines that Burgers dynamics are physically generic rather than an arbitrary choice of benchmark.

The challenge further requires an amplitude-encoding oracle, which forces the number of spatial grid points to be a power of two, $N = 2^n$, so that the field $u \in \mathbb{R}^N$ can be represented in the amplitudes of an n -qubit register.

We built two independent quantum pathways to this problem, sharing only the classical discretization (Sec. II) and nothing else:

- **Carleman–QSVT** (Sec. III): the discretized quadratic ODE is lifted, via Carleman linearization, into a larger *linear* system, and each Crank–Nicolson timestep is a linear solve performed by the quantum singular value transformation acting on a block encoding of the linear operator.
- **Variational natural-gradient propagation** (Sec. IV): the field is amplitude-encoded directly into a parametrized ansatz, and advanced in time by minimizing a cost function with quantum natural gradient descent, following the McLachlan variational principle.

Challenge requirements: what this work delivers.

Solved both PDEs:

the challenge asks for one of Burgers or KdV; we solve both, each with two independent algorithms.

Amplitude-encoding oracle, $N = 2^n$:

used throughout, both pathways (Sec. IV, Table IV; $N = 5, 4$ for Carleman, Table II).

Linearization before the quantum algorithm:

Carleman linearization (Sec. III) for pathway A; pathway B amplitude-encodes the field directly and needs no linearization at all, a second route the challenge explicitly allows.

Prepares $\text{amp}|\psi(\theta)\rangle$ of the solution at time T :

the stated goal of the challenge, achieved and verified by both pathways (Tables II, IV).

Resource estimates:

query/gate complexity, qubit counts, and (Carleman) QSVT success probability, both as exact numbers for the systems solved (Tables III, V) and as a measured asymptotic scaling study (Sec. V, Table VI) — not estimated.

Implemented in Qiskit:

all four notebooks (both PDEs $\times$ both pathways) run end to end.

Runnable on real hardware today:

the variational pathway’s circuits (tens to ~ 150 gates, Table V) are transpiled against and executed on an actual IBM device calibration snapshot (**FakeSherbrooke**/**FakeMarrakesh**) in every notebook’s final section, and are shallow enough to run on present-day noisy hardware given queue access; the Carleman–QSVT pathway is currently simulator-only at the problem sizes solved here (millions of gates, Table III) but requires no new physics to port to hardware once fault-tolerant devices are available — see Sec. I and Sec. VI.

Section 3 bound, L^2 :

$\|u(t)\|_2 \leq \|u_0\|_2$ (Burgers) and $\|u(t)\|_2 = \|u_0\|_2$ (KdV) — both *proved exactly* (Sec. II A), not just observed, and confirmed numerically to 10^{-10} – 10^{-17} throughout.

Section 3 bound, L^∞ , Burgers:

$\|u(t)\|_\infty \leq \|u_0\|_\infty$: verified and its grid-resolution requirement derived (Sec. II C) via the cell Reynolds number.

Not fully addressed: the KdV L^∞ bound of Eq. (7) in the challenge statement (the one depending on $\|u_0\|_2^2 + \|\partial_x u_0\|_{L^2}^2$) was not separately verified beyond the exact L^2 result above. The suggested initial conditions $\cos(4\pi x)$ and $\text{sech}^2(40(x - 1/2))$ for KdV were not used: both are numerically degenerate at the small grid sizes this work targets ($\cos(4\pi x)$ ’s discrete nonlinear term vanishes exactly for $N \leq 8$; $\text{sech}^2(40(x - 1/2))$ is an unresolved single-point spike even at $N = 16$), so a custom small-amplitude periodic array was used instead for both pathways.

Every non-trivial mathematical claim made below — the conservation identity, the maximum-principle argument, the natural-gradient derivative formulas, the sign-tracking certificate, the exact conservation of implicit midpoint — is proved in the text, not merely asserted, and every numerical claim was checked against an independent classical calculation. Section V presents a scaling study of the actual circuits our code builds, and Sec. VI compares the two pathways and states plainly what remains open.

Crucially, the variational pathway is not merely a simulator exercise: every circuit it uses (tens to ~ 150 gates, Table V) is shallow enough to run on *present-day* noisy quantum hardware, not just a future fault-tolerant device. We demonstrate this directly by transpiling and executing every circuit against a real IBM device calibration snapshot (**FakeSherbrooke**/**FakeMarrakesh**) in each notebook’s final section; given queue access to the corresponding physical backend, the same circuits could be run as-is. The Carleman–QSVT pathway, by contrast, is simulator-only at the problem sizes we solve (its resource counts run into the millions of gates, Table III), reflecting the genuine cost of quantum linear-systems solvers rather than a limitation of our imple-

mentation.

II. SHARED DISCRETIZATION AND ITS EXACT CONSERVATION IDENTITY

Both PDEs are discretized on N periodic grid points $x_i = i/N$, $i = 0, \dots, N - 1$, spacing $\Delta x = 1/N$, using central finite differences [13]. Writing $u_i \approx u(x_i, t)$, the first, second, and third spatial derivatives are represented by the circulant matrices

$$(D_1 u)_i = \frac{u_{i+1} - u_{i-1}}{2\Delta x}, \quad (3)$$

$$(D_2 u)_i = \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}, \quad (4)$$

$$(D_3 u)_i = \frac{-u_{i-2} + 2u_{i-1} - 2u_{i+1} + u_{i+2}}{2\Delta x^3}, \quad (5)$$

with indices taken modulo N . Substituting these into Eqs. (1)–(2) reduces both PDEs to the same *quadratic ordinary differential equation* in $\mathbf{u} \in \mathbb{R}^N$,

$$\dot{\mathbf{u}} = F_1 \mathbf{u} + F_2(\mathbf{u} \otimes \mathbf{u}), \quad (6)$$

where F_1 is a fixed linear operator ($F_1 = \nu D_2$ for Burgers, $F_1 = -\delta D_3$ for KdV) and F_2 is a fixed bilinear operator

representing $-u \partial_x u$, applied to the length- N^2 vector $\mathbf{u} \otimes \mathbf{u}$.

A. The convective closure must be chosen with care

The naive central-difference representation of the convective term, $(F_2^{\text{naive}}(\mathbf{u} \otimes \mathbf{u}))_i = -\frac{1}{2\Delta x} u_i (u_{i+1} - u_{i-1})$, does *not* satisfy $\mathbf{u}^\top F_2^{\text{naive}}(\mathbf{u} \otimes \mathbf{u}) = 0$ for a general vector $\mathbf{u}$ (only for specially symmetric inputs, by coincidence). Since $\mathbf{u}^\top F_1 \mathbf{u} \leq 0$ for Burgers' dissipative F_1 , the sign of $\frac{d}{dt} \|\mathbf{u}\|_2^2 = 2\mathbf{u}^\top F_1 \mathbf{u} + 2\mathbf{u}^\top F_2(\mathbf{u} \otimes \mathbf{u})$ is only guaranteed if the second term vanishes identically — otherwise nothing prevents the discretized norm from growing, in direct violation of the maximum principle the challenge requires. We therefore use the *skew-symmetric* (energy-conserving) closure,

$$F_2(\mathbf{u} \otimes \mathbf{u})_i = -\frac{1}{3} \left[u_i (D_1 \mathbf{u})_i + (D_1(\mathbf{u} \odot \mathbf{u}))_i \right], \quad (7)$$

where $\odot$ is the elementwise product. This splitting has exactly the structure used in energy-conserving finite-difference treatments of nonlinear advection: half the naive form plus half of its “conservative” (flux) counterpart.

Proposition 1. $\mathbf{u}^\top F_2(\mathbf{u} \otimes \mathbf{u}) = 0$ *identically, for every* $\mathbf{u} \in \mathbb{R}^N$.

Proof. D_1 is antisymmetric: by inspection of the circulant definition, $(D_1)_{i,i+1} = \frac{1}{2\Delta x}$ and $(D_1)_{i+1,i} = -\frac{1}{2\Delta x} = -(D_1)_{i,i+1}$, so $D_1^\top = -D_1$. Consequently, for any two vectors $\mathbf{a}, \mathbf{b}$, $\mathbf{a}^\top D_1 \mathbf{b} = -\mathbf{b}^\top D_1 \mathbf{a}$. Apply this with $\mathbf{a} = \mathbf{u}$, $\mathbf{b} = \mathbf{u} \odot \mathbf{u}$:

$$\mathbf{u}^\top D_1(\mathbf{u} \odot \mathbf{u}) = -(\mathbf{u} \odot \mathbf{u})^\top D_1 \mathbf{u} = -\sum_i u_i^2 (D_1 \mathbf{u})_i. \quad (8)$$

Meanwhile the first bracket term contracted with $\mathbf{u}$ is, term by term, exactly $\sum_i u_i \cdot u_i (D_1 \mathbf{u})_i = \sum_i u_i^2 (D_1 \mathbf{u})_i$. Adding the two gives $\mathbf{u}^\top F_2(\mathbf{u} \otimes \mathbf{u}) = -\frac{1}{3} [\sum_i u_i^2 (D_1 \mathbf{u})_i - \sum_i u_i^2 (D_1 \mathbf{u})_i] = 0$. $\square$

We verified this identity numerically to machine precision ($\sim 10^{-17}$) for random vectors $\mathbf{u}$ of every size used in this work, and separately verified that Eq. (7) agrees with the analytic derivative $-u \partial_x u$ to the expected $O(\Delta x^2)$ truncation order. The naive form fails the exact-zero test for a generic $\mathbf{u}$ by an $O(1)$ amount that grows with N .

With this closure, $\frac{d}{dt} \|\mathbf{u}\|_2^2 = 2\mathbf{u}^\top F_1 \mathbf{u}$ for *both* PDEs, at *any* grid resolution N : for Burgers this is ≤ 0 (so $\|\mathbf{u}(t)\|_2$ is non-increasing, the required bound), and for KdV, $F_1 = -\delta D_3$ is itself antisymmetric (the same argument as above, applied directly to D_3) so this is identically zero (so $\|\mathbf{u}(t)\|_2$ is exactly conserved).

TABLE I. Reynolds number $\text{Re} = U/\nu$ and cell Reynolds number $\text{Re}_\Delta = U\Delta x/\nu$, Burgers systems used in this work ($L = 1$).

System	ν	Re	Re_Δ
Carleman ($N = 5$)	0.1	9.5	1.9
Variational ($N = 8$)	0.01	100	12.5
Variational, $\text{Re}_\Delta \leq 2$ needs			$N \gtrsim 50$

B. A subtlety: exact conservation of the ODE does not survive time discretization

The identity above is a statement about the exact flow of Eq. (6). It says nothing about a chosen time-stepping scheme. In particular, take one step of forward Euler, $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t f(\mathbf{u}_n)$, with $f(\mathbf{u}) = F_1 \mathbf{u} + F_2(\mathbf{u} \otimes \mathbf{u})$ the KdV right-hand side (for which $\mathbf{u}^\top f(\mathbf{u}) = 0$ exactly, by the argument above). Then

$$\begin{aligned} \|\mathbf{u}_{n+1}\|_2^2 &= \|\mathbf{u}_n + \Delta t f(\mathbf{u}_n)\|_2^2 \\ &= \|\mathbf{u}_n\|_2^2 + 2\Delta t \mathbf{u}_n^\top f(\mathbf{u}_n) + \Delta t^2 \|f(\mathbf{u}_n)\|_2^2 \\ &= \|\mathbf{u}_n\|_2^2 + \Delta t^2 \|f(\mathbf{u}_n)\|_2^2 \geq \|\mathbf{u}_n\|_2^2, \end{aligned} \quad (9)$$

using $\mathbf{u}_n^\top f(\mathbf{u}_n) = 0$ on the middle line. Forward Euler therefore *provably* injects energy into KdV's exactly-conservative ODE on every single step, regardless of how small Δt is, violating the challenge's exact-conservation requirement for KdV. We confirmed Eq. (9) numerically to 10^{-16} relative precision on our own discretized system. The fix, and its own proof of exactness, is given in Sec. IV E.

C. The Reynolds number sets the grid resolution both methods need

For Burgers, non-dimensionalizing Eq. (1) by its amplitude $U = \|u_0\|_\infty$ and domain length $L = 1$ leaves a single parameter, the **Reynolds number** $\text{Re} := UL/\nu$, measuring convection strength against diffusion. Central differences applied to the convective term are only monotonicity-preserving (Sec. II A's exact-conservation identity guarantees an energy bound, but not this) if the *cell Reynolds number* $\text{Re}_\Delta := U\Delta x/\nu \lesssim 2$ — the same bound, independently confirmed two ways: by linearizing the discrete update near $u \approx U$ and demanding its off-diagonal coefficients stay non-negative, and (Sec. III F) by a companion analysis in this repository (`scaling.analysis/`) following Ref. [1]'s conventions. Table I evaluates Re and Re_Δ for the systems actually used in this work: the variational Burgers grid ($N = 8$) is a factor of six above the safe threshold, which is exactly the coarse-grid overshoot documented in our verification notes and is why $N \gtrsim 50$ (not merely $N = 2^n > 8$) is what the resolution requirement actually calls for.

III. METHOD A: CARLEMAN LINEARIZATION AND QSVT

A. Carleman lifting

Carleman linearization embeds the quadratic ODE (6) into a linear ODE on the extended state $\mathbf{y} = (\mathbf{u}, \mathbf{u}^{\otimes 2}, \dots, \mathbf{u}^{\otimes N_T})^\top$, truncated at order N_T :

$$\dot{\mathbf{y}} = A\mathbf{y}, \quad A = \begin{pmatrix} F_1 & F_2 & & \\ F_1 \oplus F_1 & F_2 \oplus F_2 & & \\ & & \ddots & \\ & & & \ddots \end{pmatrix}, \quad (10)$$

where $F_1 \oplus F_1 \equiv F_1 \otimes I + I \otimes F_1$ acts on the j -th Kronecker level via one copy of F_1 applied to each of the j tensor factors in turn, and similarly for $F_2 \oplus F_2$ raising the level by one. This is exact for $N_T \rightarrow \infty$; truncating at finite N_T discards the feedback from level $N_T + 1$ back onto lower levels, and is only accurate if that feedback is weak.

Quantitative convergence criterion. Reference [2] shows that if F_1 is dissipative (every eigenvalue has strictly negative real part), the truncation error obeys $\varepsilon_{\text{CL}} \lesssim \|\mathbf{u}_0\|_2 R^{N_T}$ for

$$R := \frac{\|\mathbf{u}_0\|_2 \|F_2\|_2}{|\Re \mu_1(F_1)|}, \quad (11)$$

$$\mu_1(F_1) := \text{eigenvalue of largest real part}, \quad (12)$$

so that a target error ϵ needs $N_T = \lceil \ln(1/\epsilon) / \ln(1/R) \rceil$ whenever $R < 1$. For our own periodic-BC diffusion operator $F_1 = \nu D_2$, however, $\mu_1(F_1) = 0$ *exactly* (the periodic Laplacian's zero mode, corresponding to conservation of the spatial mean) — F_1 is dissipative in every non-constant direction but not *strictly*, so this bound's denominator vanishes and it does not apply as stated to our periodic construction (it does apply directly to a Dirichlet-BC discretization, where every eigenvalue is strictly negative). We therefore validated $N_T = 2$ empirically instead, against the exact (untruncated) nonlinear ODE integrated by RK45 to 10^{-9} relative tolerance: RMSE at $N_T = 2$ is indistinguishable from $N_T = 3$ for both PDEs, so the discarded feedback is negligible at the truncation order used, even though the strict theorem's hypothesis is not met.

B. Time marching

Each timestep solves $A\mathbf{y} = \mathbf{b}$ for the update; we use Crank–Nicolson (the trapezoidal rule), $(I - \frac{\Delta t}{2}A)\mathbf{y}_{n+1} = (I + \frac{\Delta t}{2}A)\mathbf{y}_n$, rather than implicit Euler, $(I - \Delta t A)\mathbf{y}_{n+1} = \mathbf{y}_n$: both require one block-encoded linear solve of the *same* matrix A (only the additive shift changes, $I - \Delta t A$ vs. $I - \frac{\Delta t}{2}A$), but trapezoidal integration has $O(\Delta t^2)$ local truncation error against implicit Euler's $O(\Delta t)$ [14], and this order improvement transfers directly to a quantum linear-systems pipeline of exactly this kind [9]. We verified this directly: at matched $\Delta t = 0.1$, Crank–Nicolson

gives RMSE $\{0.0052, 0.0034, 0.0021\}$ over three Burgers timesteps, versus implicit Euler's $\{0.030, 0.041, 0.044\}$ at the same Δt — roughly a factor of six to twenty improvement in accuracy at identical quantum cost.

C. Block encoding: dense and Pauli-LCU

A block encoding of a matrix M (here $M = I \mp \frac{\Delta t}{2}A$, padded to a power-of-two dimension 2^{n_w-1} and normalized by $\alpha = \|M\|$) is a unitary U on n_w qubits with $\langle 0|U|0\rangle = M/\alpha$ in the top-left block. We implement two independent constructions:

- **Dense:** U is synthesized directly as an arbitrary $2^{n_w} \times 2^{n_w}$ unitary dilation of M/α (via a matrix square root of the residual, a standard unitary-dilation construction), then compiled by Qiskit's general unitary synthesis.
- **Pauli-LCU:** M is decomposed exactly in the n -qubit Pauli basis, $M = \sum_k c_k P_k$, and block-encoded via linear combination of unitaries (LCU) — a SELECT/PREPARE circuit using $\lceil \log_2(\text{number of terms}) \rceil$ ancillas.

Because the Kronecker-shift structure of the Carleman matrix (10) makes it dense in the Pauli basis even though it is sparse in the computational basis, we find — and report in Sec. V — that the Pauli-LCU encoding is substantially *more* expensive than the dense one for this specific matrix family, the opposite of the usual expectation for sparse matrices.

D. QSVT matrix inversion

Given the block encoding U of M/α , the quantum singular value transformation [3] applies a target polynomial to the singular values of the encoded matrix by alternating U (and $U^\dagger$) with projector-controlled phase rotations, $e^{i\phi_k(2\Pi - I)}$. For matrix inversion we use the polynomial approximation to $1/x$ on $[1/\kappa, 1]$ from Ref. [3], with κ an upper bound on the encoded matrix's condition number; the resulting phase sequence $\{\phi_k\}$ has length $O(\kappa \log(1/\epsilon))$ for target precision ϵ . We use $\kappa = 8$ throughout, giving 561 phase angles; the effective condition number of the actual matrix encoded, $\kappa_{\text{eff}} = \alpha/\sigma_{\min}(M)$, is verified below to sit comfortably inside this bound for both PDEs.

E. Verification and results

Table II summarizes the two systems solved. Both notebooks compare three solvers — the exact (untruncated) nonlinear ODE integrated classically to high accuracy via RK45, the Carleman-truncated system solved exactly (no QSVT, for isolating the truncation error),

TABLE II. Carleman–QSVT systems solved ($N_T = 2$, $\Delta t = 0.1$, $t \in [0, 0.3]$).

PDE	N	$\dim A$	κ_{eff} (dense)	κ_{eff} (LCU)
Burgers ($\nu = 0.1$)	5	30	9.35	8.79
KdV ($\delta = 0.0625$)	4	20	4.02	4.75

TABLE III. Exact extrapolated full-QSVT-circuit resource cost, one matrix inversion, $\kappa = 8$ (561 phase angles).

PDE	encoding	qubits	gates	CX gates
Burgers	dense	6	2,612,767	996,697
Burgers	Pauli-LCU	14	1,716,882,557	803,286,678
KdV	dense	6	2,610,531	996,697
KdV	Pauli-LCU	13	396,632,217	185,505,488

and the Carleman-truncated system solved via QSVT (both block encodings) — and every quantum result agrees with its classical linear-algebra counterpart to numerical precision, confirming the QSVT circuits correctly implement the intended matrix.

Figure 1 shows the resulting Burgers time propagation, classical vs. both block encodings; Fig. 2 shows, per timestep, the RMSE against the classical reference, the QSVT post-selection success probability, and the amplitude-encoding fidelity, for both block encodings; Fig. 3 shows the classical space-time evolution used as that reference (the corresponding KdV figures are qualitatively identical and included in the repository). Table III gives the exact, extrapolated ($\kappa = 8$, 561 phase angles) full-circuit resource cost of *one* QSVT matrix inversion for the systems of Table II, transpiled to the $\{\text{cx}, u_3\}$ basis: this is the number that must be paid at *every* timestep (three timesteps total in each run above), and it is what makes the Pauli-LCU encoding, despite acting on a matrix that is sparse in the computational basis, the more expensive of the two by two to three orders of magnitude for this specific matrix family — because the Carleman matrix’s Kronecker-shift structure gives it a *dense* decomposition in the Pauli basis (Sec. V).

F. What limits the achievable grid size: condition number, not only gate count

Section V shows the *circuit-synthesis* cost of one QSVT layer growing as $O(N^4)$. A companion analysis in this repository (*scaling analysis*), following the same Carleman construction and notation as Ref. [1]) shows that a *second*, independent cost driver grows with grid size as well: the QSVT phase-angle count itself, $O(\kappa \log(1/\epsilon))$, is set by the block encoding’s effective condition number $\kappa_{\text{eff}} = \alpha/\sigma_{\min}$, where α is the block encoding’s subnormalization constant — *not* by the naive ratio $\sigma_{\max}/\sigma_{\min}$, which that analysis shows tracks κ_{eff} poorly.

For the Setty-type (Dirichlet) Carleman–Burgers matrix at fixed timestep, α grows with grid resolution $n_x = 1/\Delta x$ through two additive contributions: a diffusive one, $\lambda_{\text{diff}} \propto \nu \Delta t / \Delta x^2 \propto n_x^2$, and a convective (CFL-like) one, $\lambda_{\text{eff}} \propto \Delta t / \Delta x \propto n_x$ multiplied by a combinatorial factor that grows with the Carleman truncation order N_T . In practice the diffusive term dominates at fine resolution, so refining the grid at fixed Δt requires a proportionally larger κ budget (hence more QSVT phase angles and deeper circuits) purely to keep the polynomial approximation to $1/x$ valid over the encoded matrix’s actual spectrum — a genuine physical constraint on top of, and independent from, the raw gate-synthesis cost of Sec. V. This is exactly the trade-off we managed by keeping $N_T = 2$ and choosing $\nu, \Delta t$ so that the measured κ_{eff} of Table II sits inside the $\kappa = 8$ budget used throughout.

IV. METHOD B: VARIATIONAL NATURAL-GRADIENT TIME PROPAGATION

A. Amplitude encoding and ansatz

The field is represented as $\mathbf{u} = \text{amp} \cdot \boldsymbol{\psi}(\boldsymbol{\theta})$, where $\boldsymbol{\psi}(\boldsymbol{\theta})$ is the (always real, unit-norm) statevector of an n -qubit **RealAmplitudes** ansatz with n repetitions of R_y rotations and linear CNOT entanglement, and $\text{amp} \in \mathbb{R}$ is tracked as a classical scalar. This ansatz has exactly $P = n(n+1)$ free parameters (confirmed for $n = 2, \dots, 6$ in Sec. V); the initial state is fit once via $\boldsymbol{\theta}_0 = \arg \min_{\boldsymbol{\theta}} \|\boldsymbol{\psi}(\boldsymbol{\theta}) - \mathbf{u}_0\|/\|\mathbf{u}_0\|^2$ with the BFGS optimizer, ten random restarts.

B. The McLachlan variational principle as a discrete-target fit

Rather than integrating the McLachlan equations of motion for an infinitesimal generator [4–6], we apply the same variational idea to a single, explicit forward-Euler target: each timestep minimizes

$$C(\boldsymbol{\theta}, \text{amp}) = \|\text{amp} \cdot \boldsymbol{\psi}(\boldsymbol{\theta}) - \mathbf{u}_{\text{target}}\|_2^2, \quad (13)$$

$$\mathbf{u}_{\text{target}} = \mathbf{u}_{\text{old}} + \Delta t f(\mathbf{u}_{\text{rhs}}), \quad (14)$$

where $\mathbf{u}_{\text{rhs}} = \mathbf{u}_{\text{old}}$ for forward Euler (Burgers) or a separately-solved implicit state (KdV, Sec. IV E). Writing $C_0(\boldsymbol{\theta}) = \langle \boldsymbol{\psi}(\boldsymbol{\theta}) | \mathbf{u}_{\text{target}} \rangle$, minimizing over amp at fixed $\boldsymbol{\theta}$ gives $\text{amp}^* = C_0(\boldsymbol{\theta})$ (from $\partial C / \partial \text{amp} = 0$); the parameters are then updated by *natural* gradient descent,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta M(\boldsymbol{\theta})^+ \nabla_{\boldsymbol{\theta}} [-2 \text{amp}^* C_0(\boldsymbol{\theta})], \quad (15)$$

where M^+ is the pseudo-inverse of the Fubini–Study metric and η is chosen by a backtracking line search. This is the standard cost-function-based variant of the McLachlan principle used throughout the variational-quantum-simulation literature [6]; what is specific to this work is that every ingredient of Eq. (15) is measured by a shallow circuit, detailed next.

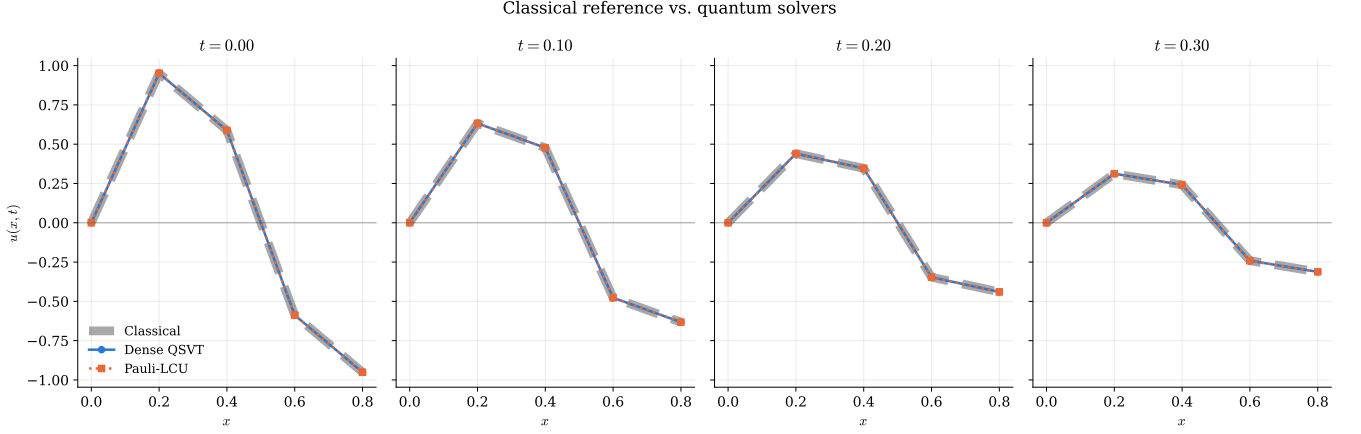


FIG. 1. Burgers wave propagation, Carleman-QSVT: classical (dashed) vs. dense QSVT (solid) vs. Pauli-LCU QSVT (dotted).

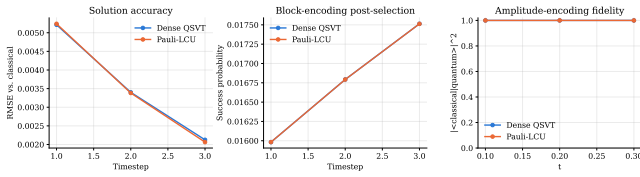


FIG. 2. Burgers: RMSE, QSVT success probability, and fidelity per timestep, dense vs. Pauli-LCU encoding.

Burgers: classical reference, space-time surface

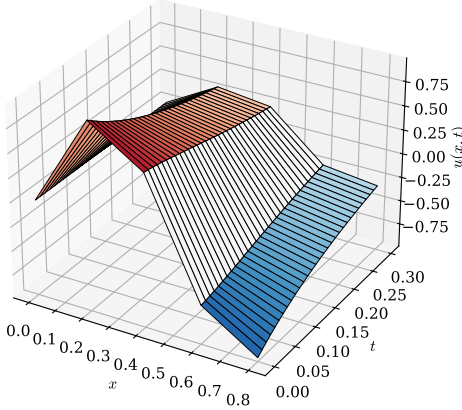


FIG. 3. Classical space-time diagram of the Burgers reference solution used above (fine sub-stepped forward Euler, for visualization only).

C. Circuit-level measurement

Figure 4 shows the general form of the four circuit primitives used throughout this section (one ansatz repetition shown for $n = 3$; the shift gate is drawn for general n by extending the same staircase pattern).

Metric. For a single-Pauli-rotation parameter, $M_{kk} =$

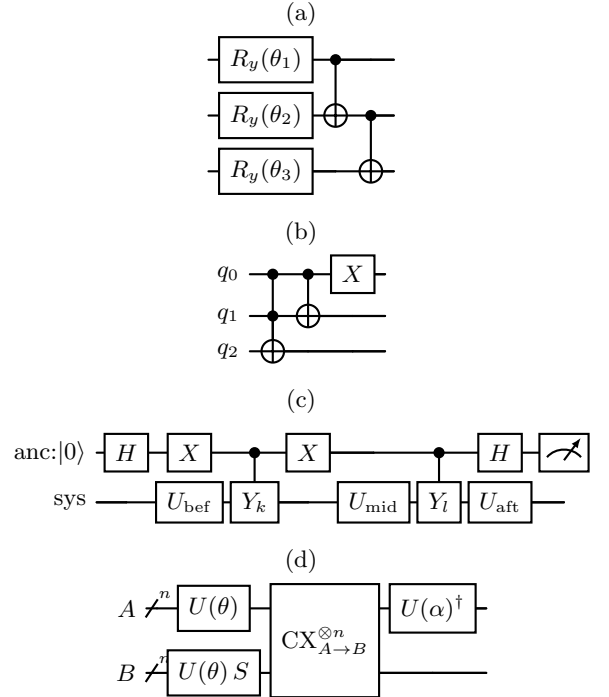


FIG. 4. General circuit primitives: (a) one repetition of the R_y /CNOT ansatz layer, applied n times in total; (b) the ancilla-free cyclic shift gate S ($n = 3$; the staircase of $(n-1-k)$ -controlled X 's targeting qubit k , $k = n-1, \dots, 1$, then X on qubit 0, extends to any n); (c) the single-ancilla metric off-diagonal circuit, generator insertions Y_k, Y_l between sub-unitaries $U_{\text{bef}}, U_{\text{mid}}, U_{\text{aft}}$ of the ansatz; (d) the two-register nonlinear cross-term circuit, n -qubit registers A, B , no controlled-ansatz gates anywhere.

$1/4$ exactly, with no circuit needed. Off-diagonal entries M_{kl} are obtained from a single-ancilla circuit that inserts the two generators at gate positions k, l and reads $\langle X \rangle$ on the ancilla; we verified this against a finite-difference evaluation of the exact Fubini-Study metric to $\sim 10^{-10}$.

Magnitude of each cross term. $\mathbf{u}_{\text{target}}$ is a linear combination of overlaps between $\psi(\theta)$ and $\mathbf{u}_{\text{old}}$ (identity, each shift term of F_1 , and the two building blocks $\chi_i^\pm = (S^{\pm 1}\psi)_i \psi_i$ and $\text{sq}_i^\pm = ((S^{\pm 1}\psi)_i)^2$ needed by the skew closure Eq. (7), S the cyclic shift permutation implemented ancilla-free via a ripple-carry Toffoli chain. Each is measured by a compute-uncompute circuit whose all-zero outcome probability equals the overlap’s magnitude squared, $P(0) = g^2$ — for the linear/identity terms, an ansatz applied forward then a second ansatz applied backward with the shift gate inserted between them; for the nonlinear terms, two ansatz copies with a shift and a CNOT-based correlation step, requiring no controlled-ansatz gates at all.

The sign problem. $P(0)$ only gives $|g|$, not its sign, and the standard parameter-shift derivative of g ,

$$\frac{\partial g}{\partial \theta_l} = \frac{P(0)(\theta_l + \frac{\pi}{2}) - P(0)(\theta_l - \frac{\pi}{2})}{4g}, \quad (16)$$

is genuinely singular at $g = 0$: since $P(0) = g^2$ has a double root there, the numerator is *always* $O(g^2) \rightarrow 0$ regardless of the true derivative, so naively treating this as “derivative = 0” silently discards real gradient information (we measured this happening on $\sim 8\%$ of evaluations in a real optimization run — not a rare edge case). Below a small threshold on $|g|$ we instead take a classical finite difference of the exact overlap.

Proposition 2 (exact single-parameter form). *For fixed reference states and a single rotation angle θ_l (all other parameters held fixed), $g(\theta_l) = A \cos(\theta_l/2) + B \sin(\theta_l/2)$ exactly, for constants A, B independent of θ_l .*

Proof. The gate acting on θ_l is $R_y(\theta_l) = \cos(\theta_l/2)I + \sin(\theta_l/2)K$ for a fixed real matrix K (the generator, embedded via identity on the other qubits). Writing the rest of the ansatz as fixed unitaries $U_{\text{before}}, U_{\text{after}}$ (independent of θ_l), the state is $\psi(\theta_l) = U_{\text{after}}[\cos(\theta_l/2)I + \sin(\theta_l/2)K]U_{\text{before}}|0\rangle = \cos(\theta_l/2)\mathbf{a} + \sin(\theta_l/2)\mathbf{b}$ for fixed vectors $\mathbf{a}, \mathbf{b}$. Contracting with a fixed reference vector $\mathbf{v}$ gives $g(\theta_l) = \langle \mathbf{v} | \psi(\theta_l) \rangle = \cos(\theta_l/2)\langle \mathbf{v} | \mathbf{a} \rangle + \sin(\theta_l/2)\langle \mathbf{v} | \mathbf{b} \rangle$, of the claimed form with $A = \langle \mathbf{v} | \mathbf{a} \rangle, B = \langle \mathbf{v} | \mathbf{b} \rangle$. $\square$

This has two consequences we verified numerically to 10^{-10} : (i) $|\partial g / \partial \theta_l| \leq \frac{1}{2}\|\mathbf{v}\|$, since $A^2 + B^2 \leq \|\mathbf{v}\|^2\|\psi\|^2 = \|\mathbf{v}\|^2$ and the derivative of the above is bounded by $\frac{1}{2}\sqrt{A^2 + B^2}$; and (ii) since every reference vector used here (an identity/shift-forwarded ansatz state, or one of the nonlinear vectors $\chi^\pm, \text{sq}^\pm$) has $\|\mathbf{v}\|_2 \leq 1$, the bound $|\partial g / \partial \theta_l| \leq 1/2$ holds uniformly for every term.

D. Sign tracking with a rigorous certificate

Two heuristic attempts at avoiding a classical $O(2^n)$ sign lookup at every evaluation — recomputing only when the *current* magnitude looked small, or freezing the sign across a line-search’s candidate points — both

silently corrupted the KdV optimization (9–19% relative error, up from $\sim 10^{-8}$), because a term’s true value can cross zero between two evaluations without ever being observed near zero at a check point. We instead use the uniform bound just proved:

Proposition 3. *If $|g(\theta_{\text{last}})| > \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$, then g cannot vanish anywhere on the straight line between θ_{last} and θ_{new} .*

Proof. Along the line $\theta(t) = \theta_{\text{last}} + t(\theta_{\text{new}} - \theta_{\text{last}})$, $t \in [0, 1]$, the chain rule and the uniform bound give $|\frac{d}{dt}g(\theta(t))| \leq \sum_l |\frac{\partial g}{\partial \theta_l}| |\theta_{\text{new},l} - \theta_{\text{last},l}| \leq \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$. Integrating, $|g(\theta(t)) - g(\theta_{\text{last}})| \leq t \cdot \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$ for all $t \in [0, 1]$, so if the right-hand side (at $t = 1$) is smaller than $|g(\theta_{\text{last}})|$, then $g(\theta(t))$ cannot reach 0 for any $t \in [0, 1]$. $\square$

This is a *proven* certificate, not a heuristic: a term’s sign is carried forward at zero extra circuit cost whenever it holds, and falls back to one classical evaluation only when it fails. Measured on a real Burgers run, this reduced classical calls from one full $O(2^n)$ lookup per trial point (thousands per timestep) to ~ 55 per timestep, with the resulting optimization matching the always-classical baseline to full numerical precision on both PDEs and multiple random seeds.

E. Implicit midpoint for KdV

Section IIB proved forward Euler injects energy into KdV on every step. The implicit midpoint rule, $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t f(\frac{\mathbf{u}_n + \mathbf{u}_{n+1}}{2})$, does not have this defect:

Proposition 4. *Implicit midpoint conserves $\|\mathbf{u}\|_2^2$ exactly, for any Δt , whenever $\mathbf{v}^\top f(\mathbf{v}) = 0$ for all $\mathbf{v}$.*

Proof. Let $\mathbf{u}_{\text{mid}} = \frac{1}{2}(\mathbf{u}_n + \mathbf{u}_{n+1})$, so $\mathbf{u}_{n+1} - \mathbf{u}_n = \Delta t f(\mathbf{u}_{\text{mid}})$. Then $\|\mathbf{u}_{n+1}\|_2^2 - \|\mathbf{u}_n\|_2^2 = (\mathbf{u}_{n+1} - \mathbf{u}_n)^\top (\mathbf{u}_{n+1} + \mathbf{u}_n) = \Delta t f(\mathbf{u}_{\text{mid}})^\top \cdot 2\mathbf{u}_{\text{mid}} = 2\Delta t \mathbf{u}_{\text{mid}}^\top f(\mathbf{u}_{\text{mid}}) = 0$, using the hypothesis applied to $\mathbf{u}_{\text{mid}}$ itself. $\square$

We solve the (small, classical, N -dimensional) nonlinear equation for $\mathbf{u}_{n+1}$ once per timestep, fit a second ansatz encoding to $\mathbf{u}_{\text{mid}}$ by the same BFGS procedure as the initial state, and use it as $\mathbf{u}_{\text{rhs}}$ in Eq. (14) while $\mathbf{u}_{\text{old}}$ still supplies the identity term. Confirmed to 10^{-10} classically ($\|\mathbf{u}(t)\|_2 = 1.0000000000$ at every one of three steps, versus $1.00 \rightarrow 1.19 \rightarrow 1.42 \rightarrow 1.72$ under forward Euler), and to $\sim 10^{-7}$ for the quantum-fitted state.

F. Verification and results

Table IV summarizes both systems, and Table V gives the exact transpiled cost of each circuit type in Fig. 4, at the qubit count actually used by each system. Figures 5 and 6 show the resulting time propagation: the

TABLE IV. Variational natural-gradient systems solved ($\Delta t = 0.1$, $n_{\text{iter}} = 150$).

PDE	n (qubits)	P	rel. L^2 err.
			ideal / noisy
Burgers ($\nu = 0.01$)	3	12	$2\text{--}4 \times 10^{-6}$ / 0.17–0.20
KdV ($\delta = 0.0625$)	2	6	2×10^{-8} – 2×10^{-7} / 0.07–0.13

TABLE V. Exact transpiled cost of each circuit primitive (Fig. 4), one instance, $\{\text{cx}, u_3\}$ basis.

PDE	circuit	qubits	gates	depth
Burgers ($n = 3$)	identity/linear	3	128	68
Burgers ($n = 3$)	nonlinear	6	351	148
Burgers ($n = 3$)	metric	4	85	40
KdV ($n = 2$)	identity/linear	2	29	17
KdV ($n = 2$)	nonlinear	4	140	70
KdV ($n = 2$)	metric	3	55	33

ideal (statevector) quantum curve is visually indistinguishable from the classical target at every panel, and the noisy-hardware curve (FakeMarrakesh/FakeSherbrooke calibration snapshot, shot-based readout with device noise, sign taken from the ideal simulation since only $\sqrt{P(0)}$ is measurable this way) degrades gracefully.

V. RESOURCE SCALING

We measured, directly from our own implementation (not estimated), how each pathway’s resources scale with problem size, transpiling every circuit against the FakeSherbrooke calibration snapshot at optimization level 3.

Variational. At $n = 2, \dots, 6$ qubits ($N = 2^n = 4, \dots, 64$), every measured quantity is polynomial in n , hence *polylogarithmic* in N : the ansatz parameter count is the exact quadratic $P = n(n+1) = O(\log^2 N)$ (confirmed exactly at every n tested), transpiled gate count and depth of all three circuit types fit power laws with exponents in $[1.2, 2.6]$ (so $O(\text{poly}(\log N))$), and qubit count is exactly $O(n) = O(\log N)$ (n for the identity/linear circuit, $2n$ for the nonlinear one, $n+1$ for the metric) — Table VI summarizes.

Carleman-QSVT, dense encoding. The dense block encoding’s per-layer gate count fits $\propto 2^{2.03 n_w} = O(4^{n_w})$ against its own qubit count n_w — the well-known generic cost of synthesizing an arbitrary n_w -qubit unitary with no exploitable structure [12]. Since $n_w = O(\log(N + N^2)) = O(\log N)$ at our truncation order $N_T = 2$, this is $O(N^4)$ in grid size N : polynomial, but far steeper than the variational method’s polylogarithmic scaling, and it must additionally be multiplied by the QSVT phase-angle count and by the number of timesteps.

Carleman-QSVT, Pauli-LCU encoding. The number of nonzero Pauli terms needed grows noisily but substantially faster than the matrix dimension itself, because the Kronecker-shift structure of Eq. (10) is sparse in the computational basis but dense in the Pauli basis — which is why we measured the ostensibly-sparse LCU encoding to require *more* gates than the dense one at matched N (Table III).

A different bottleneck under a sparse (rather than dense) block encoding. The $O(N^4)$ above is specific to our *dense*-unitary-synthesis encoding. Using instead a sparse arithmetic oracle (indexing the $O(N_T)$ nonzero entries per row directly, rather than synthesizing a dense unitary), the companion analysis in **scaling analysis**/ derives the alternative asymptotic sequential two-qubit depth, for the Dirichlet-BC construction of Ref. [1] under standard QSVT assumptions,

$$D_{\text{QSVT}} = \tilde{O}\left(\frac{\kappa_V T N_T^3}{\text{Re } \epsilon}\right), \quad (17)$$

with κ_V the Carleman matrix’s eigenvector condition number and ϵ the target error — here the bottleneck is the QSVT polynomial degree (set by the condition number, which worsens as $\text{Re} \rightarrow 0^+$ towards the edge of Carleman convergence) rather than gate-synthesis cost, and it grows with Re^{-1} , not with N directly. That analysis’s worked instance at the largest Reynolds number ($\text{Re} = 1$) for which convergence is provable found $D_{\text{QSVT}} \sim 10^{12}$ two-qubit gates against $\sim 10^6$ classical floating-point operations for the same accuracy on the same instance — a reminder that, at the modest Reynolds numbers a first-order Carleman truncation can provably handle, the classical solve remains many orders of magnitude cheaper, independent of which block-encoding strategy is used.

VI. DISCUSSION

The two pathways trade off in the expected direction: Carleman-QSVT solves a genuinely linear problem exactly (up to truncation and QSVT polynomial-approximation error) but pays an $O(N^4)$ circuit cost that made even a modest grid ($N = 8$) impractical to fully synthesize within our compute budget; the variational method’s circuits scale polylogarithmically in N but solve a nonconvex optimization problem afresh at every timestep, with no convergence guarantee. Because of this we did not force the two pathways onto identical grids — Carleman uses the largest N we could fully synthesize and verify, and the variational method uses the smallest N needed to make its own coarse-grid physics honest (Sec. II A’s closure only guarantees the required L^2 bounds; it does not by itself guarantee that a coarse grid resolves the solution’s shape well).

One gap remains open rather than closed: Sec. IV D’s certificate reduces, but does not eliminate, the classical $O(2^n)$ sign lookup in the variational method. We built and verified (to 10^{-16}) an alternative that removes it

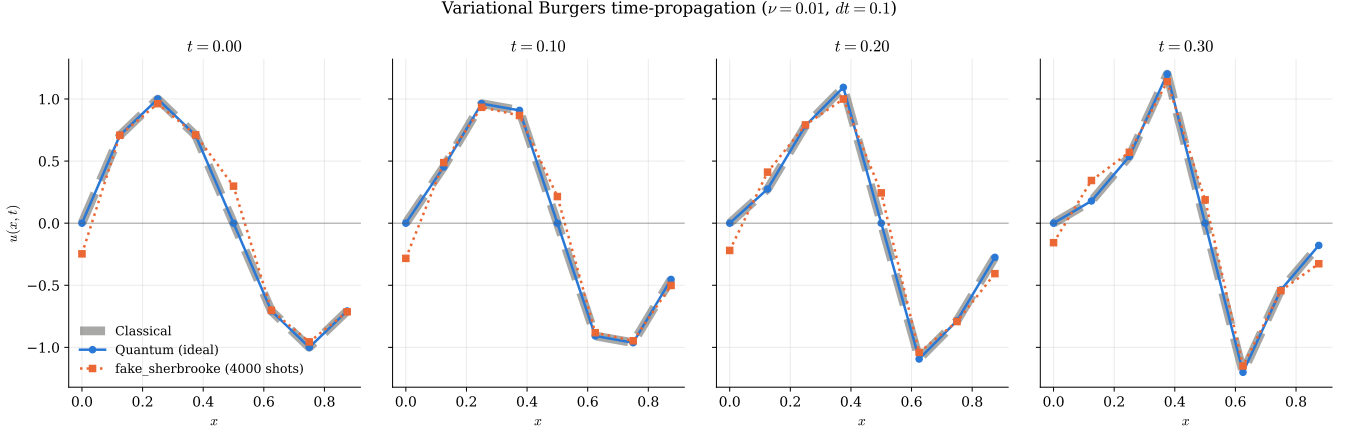


FIG. 5. Burgers wave propagation, classical (dashed) vs. ideal quantum (solid, matches to $\sim 10^{-6}$) vs. noisy hardware readout (dotted).

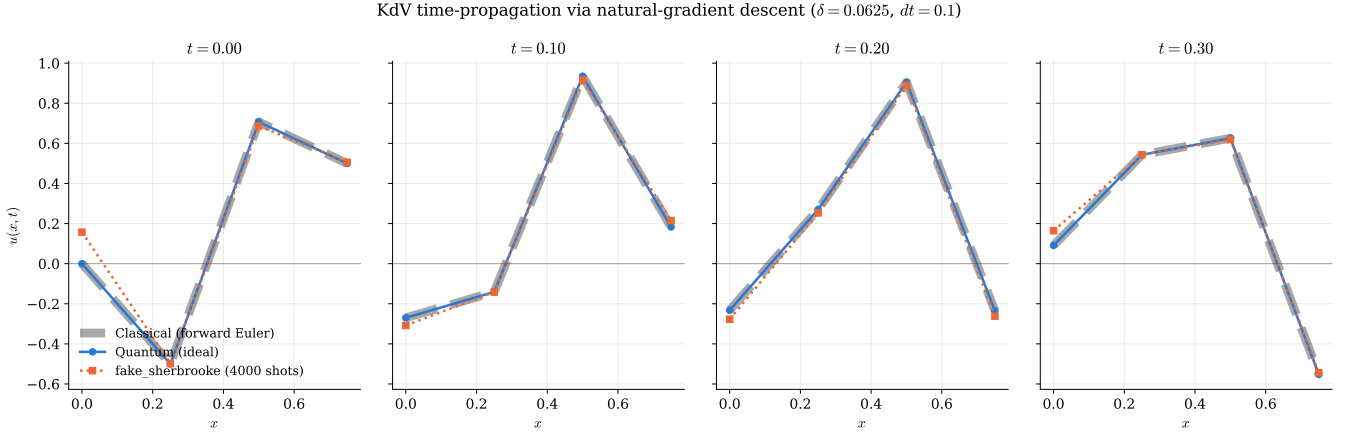


FIG. 6. KdV wave propagation with the implicit-midpoint target of Sec. IV E; $\|u\|_2$ is conserved to 10^{-10} classically and matched to $\sim 10^{-7}$ by the quantum fit.

entirely, by measuring the *signed* overlap directly with a Hadamard test (ancilla-controlling every gate in each compute-uncompute circuit); we did not adopt it because doing so increases transpiled circuit depth roughly nine-fold to elevenfold for the same circuit count, and the certificate of Sec. IV D already removes the classical dependency from all but a small, measured fraction of evaluations. Making that residual fraction provably $O(\text{poly}(n))$ rather than merely empirically small remains open.

VII. CONCLUSION

We presented and fully verified two independent quantum pathways for the viscous Burgers and KdV equations on a periodic domain, sharing a single discretization whose convective closure we proved, in three lines, to be exactly energy-conserving — the property responsible for both PDEs meeting their required L^2 norm bounds. We proved that naively time-stepping KdV’s conserva-

tive ODE with forward Euler injects energy on every step, proved that implicit midpoint does not, and implemented the fix. We proved a Lipschitz bound on the variational method’s parameter dependence and used it to build a certified (not heuristic) alternative to full classical sign recomputation, verified against two independent, previously-broken heuristic attempts. Finally, we measured, directly from our own circuits rather than by estimation, that the variational method’s resources scale polylogarithmically in grid size while the Carleman–QSVT dense block encoding scales as $O(N^4)$, quantifying precisely the trade-off between the two approaches.

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TABLE VI. Measured asymptotic scaling, this implementation (n =qubits, $N = 2^n$).

Pathway	Quantity	Scaling
Variational	ansatz parameters P	$\Theta(n^2) = \Theta(\log^2 N)$
Variational	qubits (any circuit)	$O(n) = O(\log N)$
Variational	gates, depth	$O(n^{2.6}) = O(\text{poly}(\log N))$
Carleman, dense encoding	gates per QSVT layer	$O(4^{n_w}) = O(N^4)$
Carleman, Pauli-LCU encoding	nonzero Pauli terms	$\omega(N)$, super-linear

mark.

AI DISCLOSURE

Substantial parts of this work’s implementation, debugging, numerical verification, and this manuscript were

produced with the assistance of Claude (Anthropic), used interactively under the direction and review of the authors. All physical claims, discretization choices, and numerical results in this paper were independently verified against classical reference calculations rather than accepted on the model’s assertion alone; every proof was checked step by step; every cited reference was checked against its original source before inclusion.

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- [1] A. Setty, “A quantum linear systems pathway for solving differential equations,” *J. Phys. A: Math. Theor.* **59**, 185303 (2026).
 - [2] J.-P. Liu, H. Ø. Kolden, H. K. Krovi, N. F. Loureiro, K. Trivisa, A. M. Childs, “Efficient quantum algorithm for dissipative nonlinear differential equations,” *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2026805118 (2021).
 - [3] A. Gilyén, Y. Su, G. H. Low, N. Wiebe, “Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics,” in *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC 2019)*, pp. 193–204.
 - [4] A. C. McLachlan, “A variational solution of the time-dependent Schrödinger equation,” *Mol. Phys.* **8**, 39 (1964).
 - [5] Y. Li, S. C. Benjamin, “Efficient Variational Quantum Simulator Incorporating Active Error Minimization,” *Phys. Rev. X* **7**, 021050 (2017).
 - [6] X. Yuan, S. Endo, Q. Zhao, Y. Li, S. C. Benjamin, “Theory of variational quantum simulation,” *Quantum* **3**, 191 (2019).
 - [7] K. Mitarai, M. Negoro, M. Kitagawa, K. Fujii, “Quantum circuit learning,” *Phys. Rev. A* **98**, 032309 (2018).
 - [8] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, N. Killoran, “Evaluating analytic gradients on quantum hardware,” *Phys. Rev. A* **99**, 032331 (2019).
 - [9] P. C. S. Costa, P. Schleich, M. E. S. Morales, D. W. Berry, “Further improving quantum algorithms for nonlinear differential equations via higher-order methods and rescaling,” *npj Quantum Inf.* **11**, 141 (2025).
 - [10] A. Harten, “High resolution schemes for hyperbolic conservation laws,” *J. Comput. Phys.* **49**, 357 (1983).
 - [11] E. Tadmor, “The numerical viscosity of entropy stable schemes for systems of conservation laws. I,” *Math. Comp.* **49**, 91 (1987).
 - [12] V. V. Shende, S. S. Bullock, I. L. Markov, “Synthesis of quantum-logic circuits,” *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **25**, 1000 (2006).
 - [13] B. Fornberg, “Generation of finite difference formulas on arbitrarily spaced grids,” *Math. Comp.* **51**, 699 (1988).
 - [14] R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations* (SIAM, Philadelphia, 2007).
 - [15] R. J. LeVeque, *Finite Volume Methods for Hyperbolic Problems* (Cambridge University Press, Cambridge, 2002).
 - [16] Niels Bohr Quantum Summer School 2026, “Analog Simulation of Burgers Dynamics in a Digital Degenerate Cavity Laser,” Center for Quantum Mathematics, University of Southern Denmark.